

AcEvery model makes more errorscent Bias in Speech Recognition

Does OpenAI's Whisper transcribe some English accents less accurately, and does a bigger model close the gap?

Jeo Metelitsa · Project report, October 2026 · Data: Mozilla Common Voice English 27.0 · Code: github.com/jeoethecreator/accent-bias-asr

Summary. Whisper makes two to three times as many errors on South Asian, Southern African, Southeast Asian and second-language European English as it does on US, English, Canadian and Australian/New Zealand English, and this holds for every model size tested. Larger models reduce errors for every group: the largest model makes about 75% fewer errors than the smallest. But the improvement is smaller in relative terms for the groups that start out worse, so the relative gap grows with model size, from 2.1 times to 2.7 times. The difference between groups shrinks in absolute terms (from 15.7 to 4.9 percentage points) while the ratio between them increases, and the increase is statistically reliable.

At a glance

What was tested	Result
Clips transcribed	3,200 clips from 2,938 speakers in 8 accent groups (400 clips per group)
Models	6 Whisper models (tiny, base, small, medium, large-v3, turbo) and wav2vec 2.0 for comparison; 22,400 transcriptions in total
Best and worst group (large-v3)	Canada 1.4% word error rate; Philippines/Malaysia/Singapore 6.2%
Gap in percentage points (tiny to large-v3)	15.7 to 4.9 points: the absolute gap shrinks
Ratio, other four groups to inner circle (tiny to large-v3)	2.11 to 2.74 times: the relative gap grows (change +0.63, 95% CI +0.22 to +1.21)
Groups significantly worse than the US group	All four non-inner-circle groups, on all 7 models

1. Question and hypothesis

Research question. Does Whisper's word error rate (WER) differ across English accent groups in Mozilla Common Voice, and does the gap between groups change across Whisper model sizes?

Hypothesis. WER will be higher for accents outside the US and UK, and larger models will reduce overall WER but will not close the gap proportionally.

Why it matters. Speech recognition runs captions, voice assistants, call centres and accessibility tools. If it works worse for some accents, those speakers get a worse product or cannot use it at all. Koenecke et al. (2020) found that five major commercial systems made about twice as many errors for Black speakers as for white speakers. This project applies a similar audit to Whisper, an open model that is widely built into other products, and adds a question that study could not ask: what happens to the gap as the same model is made larger.

2. Data

Source

Mozilla Common Voice is a public dataset of volunteers reading sentences aloud and optionally describing their own accent. I used the English validated set from release 27.0 (September 2026), downloaded from Mozilla Data Collective. Only the metadata and the 3,200 selected audio clips were extracted from the 88 GB download, and no audio is redistributed in the project repository.

From 1.9 million clips to 3,200

Step	Clips	Speakers
Validated clips in Common Voice English 27.0	1,929,854	84,779
Has an accent label (44.8% of clips had none)	1,065,053	19,035
Label maps cleanly to one accent group	996,236	17,533
Sentence has at least 4 words	976,621	17,304
Audio at least 1 second long	976,598	17,304

Step	Clips	Speakers
Final sample	3,200	2,938

Accent groups

Accent labels in Common Voice are free text written by the speakers themselves, and English 27.0 has 868 different ones. I grouped them into 8 groups that follow Kachru's "three circles" of English: the inner circle, where English is the main native language; the outer circle, where English is widely used in education and government; and the expanding circle, where English is learned as a foreign language.

Group	Circle	What it includes	Speakers in sample
US	Inner	United States English, including regional descriptions such as Midwestern	400
England	Inner	England English, including regions such as Yorkshire or Liverpool	400
Canada	Inner	Canadian English, including provinces	400
Australia/NZ	Inner	Australian and New Zealand English, pooled	400
South Asia	Outer	India, Pakistan, Sri Lanka, Bangladesh, Nepal	400
Southern Africa	Outer	South Africa, Zimbabwe, Namibia and neighbours	353
SE Asia (PH/MY/SG)	Outer	Philippines, Malaysia, Singapore	347
L2 (European L1)	Expanding	Non-native speakers whose first language is European, such as German, French or Spanish	238

The grouping rules are an ordered list of patterns, each with a written reason, and a clip is kept only if

every part of its label points to the same group. "United States English|Midwestern" goes to US, but "United States English|England English" is dropped as a conflict. Labels that were ambiguous ("British", which could mean England, Scotland or Wales), mixed ("Mixed American and British"), or describing speech differences rather than accents were excluded on purpose. Some substrings needed special handling: "New England" is a US region, and "West Indian" refers to the Caribbean, not India. The full rule list and every label's mapping are in the data card in the repository.

How the 400 clips per group were chosen

1. **Speakers first.** Sampling runs in passes. The first pass allows one clip per speaker, the second allows two, and so on up to a cap of ten. A group only returns to a speaker once it has used every available new speaker. In the end no speaker contributed more than two clips.
2. **Shared sentences second.** Within each pass, sentences read by the most groups are taken first, so different groups read the same sentences where possible. 489 sentences appear in at least two groups.
3. **No repeats.** No sentence appears twice in the same group. The random choices use a fixed seed, so the same inputs always give the same sample.

Speaker spread was put ahead of sentence matching because clips from one person are not independent. A first version of the sampler put sentence matching first and used only 193 of the roughly 1,000 Canadian speakers available, with some speakers contributing ten clips; the final version uses 400.

What the data revealed along the way

- One speaker had recorded 53,295 clips alone, which is why the cap on clips per speaker is needed.
- 44.8% of validated clips had no accent label at all and could not be used.
- The sample is about 75% male (by self-report). The South Asia group is about 91% male, and 57% of the L2 European group left gender blank. Gender was reported but not balanced, so it is a possible confound.

3. The test

Models

Model	Parameters	Notes
Whisper tiny	39M	Smallest Whisper model

Model	Parameters	Notes
Whisper base	74M	
Whisper small	244M	
Whisper medium	769M	
Whisper large-v3	1,550M	Largest and most accurate Whisper model
Whisper turbo	809M	A pruned large-v3 with fewer decoder layers, built for speed
wav2vec 2.0 (large, 960h)	317M	Non-Whisper comparison model from Meta, trained only on LibriSpeech audiobooks

How each clip was transcribed

- Every Whisper model used identical settings: language fixed to English, beam search with 5 beams and temperature fallback (the settings used in the Whisper paper), and a fixed random seed for each clip so results can be reproduced.
- Transcription ran on a free T4 GPU in Google Colab. Each transcript was saved to Google Drive as soon as it was produced, so the run survived several disconnects without losing work. In total the runs took about 4 hours of GPU time.
- Clips were processed in a fixed shuffled order, so a partial run would still cover every group.

How errors were counted

Before scoring, both the reference sentence and the model's transcript go through Whisper's English text normalizer, so differences in punctuation, capitalization, spelling variety or number format ("Mr." and "mister", "colour" and "color") are not counted as errors. The word error rate is then:

$$\text{WER} = (\text{substitutions} + \text{deletions} + \text{insertions}) \div \text{words in the reference}$$

For example, if the reference is "what joy there is in living" and the model writes "what joy there is living", one word was deleted out of six, so WER is 1/6, or 16.7%. Group results use pooled WER, which is the total number of errors divided by the total number of reference words, so a long sentence counts more than a short one.

Quality checks

- Random transcripts from every model were read by hand before the larger models were run.
- Only 5 of the 22,400 transcriptions came back empty.
- The code has 57 automated tests, including tests that simulate a Colab disconnect in the middle of writing a result.

4. Analysis method

Confidence intervals. Every WER comes with a 95% confidence interval from a speaker-level bootstrap. Within each group, speakers are resampled with replacement 1,000 times, keeping all clips from each chosen speaker, and WER is recalculated each time. The interval is the middle 95% of those values. Resampling speakers rather than individual clips matters because two clips from the same person are not independent, and treating them as independent would make the intervals too narrow.

Measuring the gap. Two measures are used:

- **Absolute gap:** the worst group's WER minus the best group's WER, in percentage points.
- **Relative gap:** the pooled WER of the four outer- and expanding-circle groups divided by the pooled WER of the four inner-circle groups. This uses all eight groups, so it is steadier than comparing only the single best and worst group.

Because the same bootstrap resample is applied to every model, differences between models are paired, which allows a confidence interval on the change in the gap from one model to another.

5. Results

Every model makes more errors on outer- and expanding-circle accents

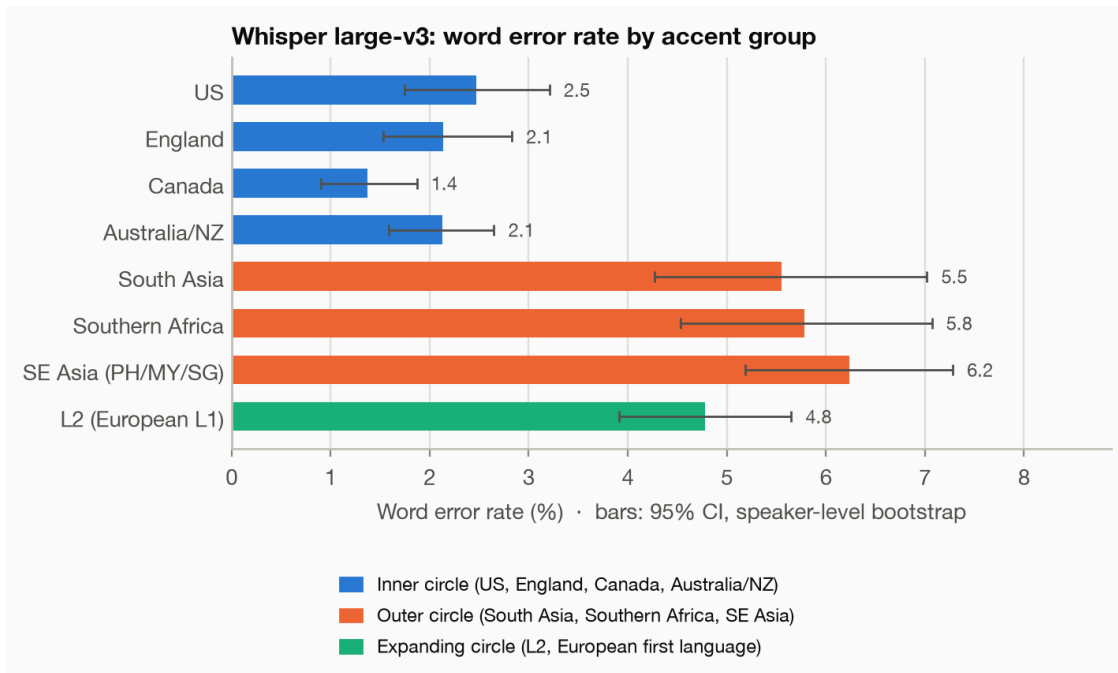


Figure 1. Word error rate by accent group for Whisper large-v3, the most accurate model. Error bars are 95% confidence intervals. The four inner-circle groups (blue) range from 1.4% to 2.5%; the other four groups range from 4.8% to 6.2%.

Group	tiny	base	small	medium	large-v3	turbo	wav2vec2
US	9.2	6.3	4.1	3.1	2.5	2.4	5.2
England	11.5	6.7	4.2	2.5	2.1	2.2	6.1
Canada	7.6	5.2	3.1	1.9	1.4	1.5	6.0
Australia/NZ	10.8	7.3	4.3	2.4	2.1	2.6	7.3
South Asia	19.8	14.7	9.7	7.1	5.6	6.8	14.7
Southern Africa	21.4	15.4	9.9	6.5	5.8	7.5	14.7
SE Asia (PH/MY/SG)	23.2	16.1	10.5	7.8	6.2	7.1	17.1
L2 (European)	18.5	13.9	9.2	6.5	4.8	5.5	13.8

Group	tiny	base	small	medium	large-v3	turbo	wav2vec2
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Table 1. Pooled word error rate (%) for every group and model. Confidence intervals for every cell are in results/wer_by_group_model.csv in the repository.

Compared with the US group, each of the four non-inner-circle groups has a higher WER on all seven models, and every one of those 28 differences has a 95% confidence interval that excludes zero. Among the inner-circle groups, England is not reliably different from the US on any model, and Australia/NZ differs only on wav2vec 2.0 (2.0 points worse). Canada is slightly better than the US on medium, large-v3 and turbo (about 1 percentage point).

Bigger models shrink the gap in points but widen it in proportion

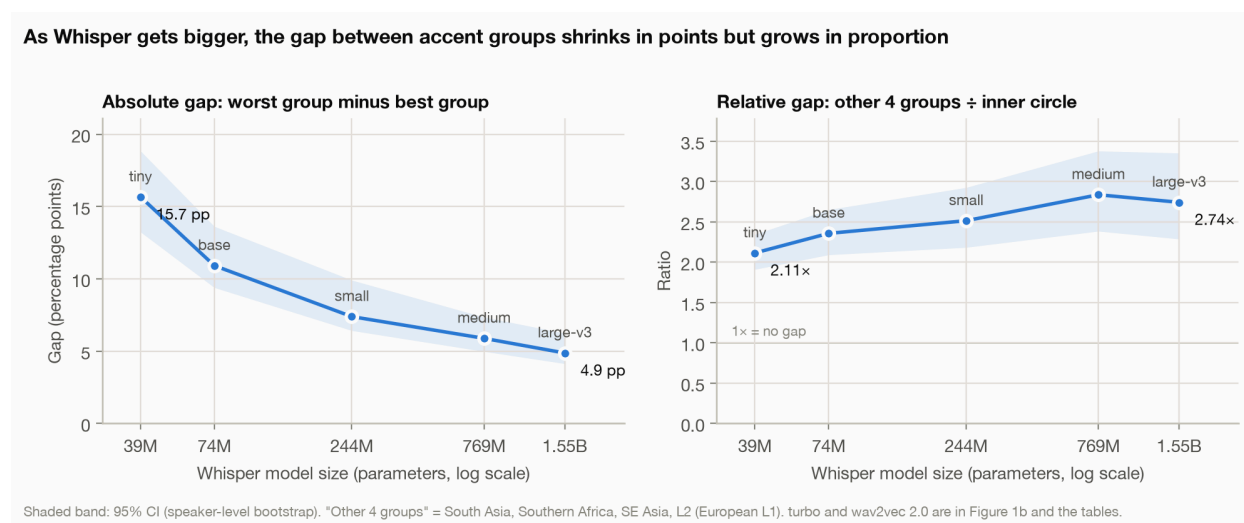


Figure 2. Left: the absolute gap between the worst and best group falls from 15.7 to 4.9 percentage points as Whisper grows from 39 million to 1.55 billion parameters. Right: over the same range, the other four groups' error rate rises from 2.11 to 2.74 times the inner circle's. Shaded bands are 95% confidence intervals.

Model	Inner circle WER	Other groups WER	Ratio (95% CI)	Absolute gap, pp (95% CI)
tiny	9.8%	20.7%	2.11 (1.91 to 2.34)	15.7 (13.2 to 18.8)
base	6.4%	15.0%	2.36 (2.09 to 2.65)	10.9 (9.4 to 13.6)
small	3.9%	9.8%	2.51 (2.18 to 2.93)	7.4 (6.4 to 9.9)
medium	2.5%	7.0%	2.84 (2.38 to 3.38)	5.9 (5.0 to 7.3)

Model	Inner circle WER	Other groups WER	Ratio (95% CI)	Absolute gap, pp (95% CI)
large-v3	2.0%	5.6%	2.74 (2.29 to 3.35)	4.9 (4.1 to 6.2)
turbo	2.2%	6.7%	3.06 (2.58 to 3.69)	6.0 (5.1 to 7.8)
wav2vec2	6.1%	15.1%	2.46 (2.17 to 2.77)	11.9 (10.0 to 14.1)

Table 2. The gap between groups for every model. "Other groups" means South Asia, Southern Africa, SE Asia and L2 European, pooled.

From tiny to large-v3, the inner circle's error rate falls by 79% (9.8% to 2.0%) and the other groups' falls by 73% (20.7% to 5.6%). Everyone benefits, but the groups that start out worse benefit less in relative terms, so the ratio between them increases. The change in the ratio from tiny to large-v3 is +0.63, with a 95% confidence interval of +0.22 to +1.21. Because the interval does not include zero, the widening is very unlikely to be caused by random variation in which speakers were sampled.

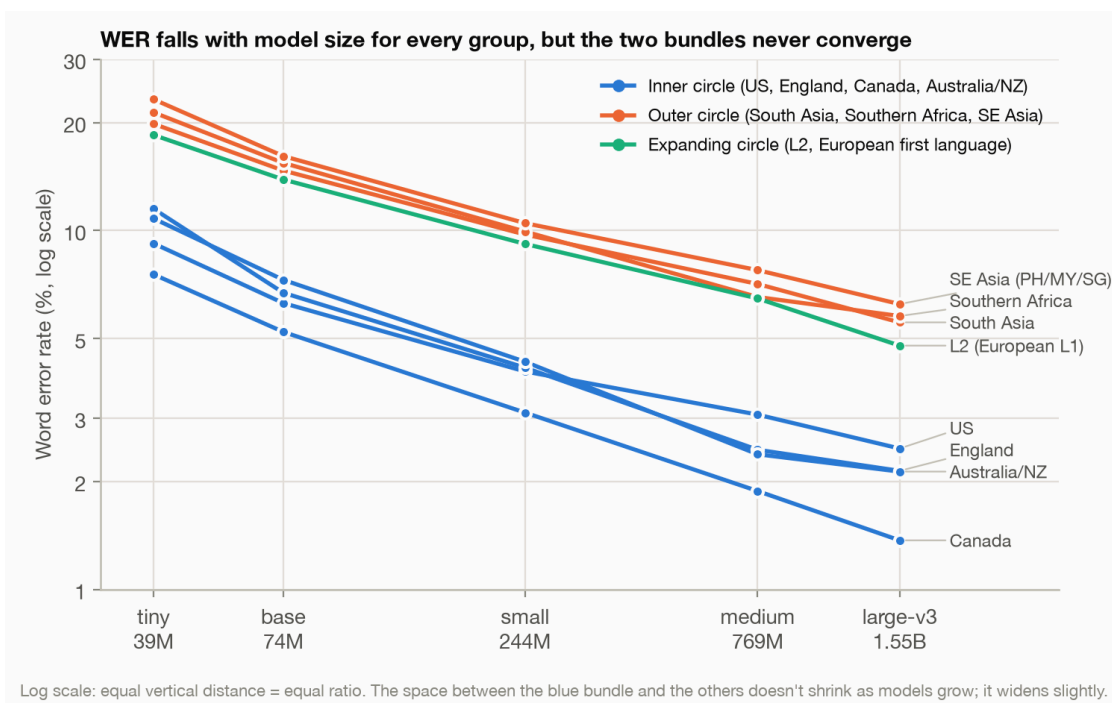


Figure 3. WER for each group across Whisper sizes, on logarithmic scales. On a log scale, equal vertical distance means an equal ratio, so the fact that the blue bundle and the other lines do not move closer together shows that the relative gap does not close.

Two further observations

- **turbo widens the gap further.** turbo is a faster, pruned version of large-v3. It is about as accurate as large-v3 for the inner circle (2.2% vs 2.0%) but less accurate for the other groups (6.7% vs

5.6%), so its ratio of 3.06 is the highest of any model. Making the model faster cost the most accuracy for the groups that were already worst served.

- **The gap is not specific to Whisper.** wav2vec 2.0, a different model trained on a different kind of data (audiobooks only), shows a ratio of 2.46, between Whisper base and small.

6. Robustness check: clips where the model failed completely

The 50 clips with the highest large-v3 WER include several hallucinations, where Whisper wrote fluent text that has nothing to do with the audio, such as "Transcription by CastingWords" or "Hello friends. Today, I will explain you...". These are a known failure of Whisper, and a single one can add many insertion errors.

To check that these clips are not driving the result, I removed the 82 clips on which any model scored a WER of 100% or more and recalculated the ratio. The ratio is a little lower without them, but the pattern is the same: it still rises from 1.95 (tiny) to 2.47 (large-v3). The removed clips come mostly from South Asia (21) and SE Asia (20), so whether they reflect the model failing on those accents or problems with the recordings is an open question, addressed in the next steps below.

Model	Ratio, all clips	Ratio without the 82 clips
tiny	2.11	1.95
base	2.36	2.18
small	2.51	2.34
medium	2.84	2.64
large-v3	2.74	2.47
turbo	3.06	2.76
wav2vec2	2.46	2.31

7. What the results mean

The hypothesis is supported, and the result is stronger than the hypothesis predicted. Larger models do reduce errors for everyone, and the absolute difference between groups shrinks. But larger models do not just fail to close the gap proportionally; they widen it. A speaker with a South Asian, Southern African, Southeast Asian or second-language accent gets about 2.1 times as many errors as an inner-circle speaker from the smallest model and about 2.7 times as many from the largest.

This matters for how improvements are reported. A headline that says a new model "cuts errors by 75%"

is true on average, but the benefit is not shared evenly, and the speakers who were worst served before end up further behind in relative terms. The turbo result shows that the same can happen when a model is made faster: the accuracy given up for speed came mostly from the groups that were already worse served.

8. Limitations

- **Self-reported, grouped labels.** Accent labels are written by the speakers themselves, and the grouping into eight groups is a judgement call. Each group also pools real variation, such as all regions of the US or all European first languages.
- **Read speech.** Common Voice speakers read prepared sentences, so gaps in natural conversation may be different.
- **Who contributes.** Common Voice contributors are not a random sample of each accent's speakers. The sample is about 75% male, and the gender balance differs between groups.
- **Recording conditions.** Speakers record on their own devices, and microphone and background-noise quality may differ between groups.
- **Sample size.** 400 clips per group gives confidence intervals of roughly plus or minus 1 to 3 percentage points; smaller differences cannot be detected.
- **Scope.** The results apply to the specific models tested, not to all speech recognition.

9. Ethics

- Results are reported only for groups, never for individuals. Common Voice speaker IDs were replaced with project-only IDs, and no attempt was made to identify anyone.
- Common Voice audio is not redistributed. The repository contains clip IDs, code and processing steps, so others can download the data and reproduce the work themselves.
- Findings describe where the system underperforms, not how anyone should speak.

10. Next steps

1. **Listen to and tag the 50 worst clips** (results/worst_clips_to_tag.csv), labelling each failure as a hallucination, a name or rare word, specific sounds, speaking speed, background noise, or a label error. This will show whether the extreme failures come from accents or from recording problems.
2. **Check the gender confound** by comparing WER for male and female speakers within each group.
3. **Write the paper** (Phase 5): abstract, introduction, related work, methods, results, limitations and

conclusion, followed by a poster and a short plain-language post.

11. How to reproduce this

All code is at github.com/jeothecreator/accent-bias-asr, with automated tests and a step-by-step README.

Phase	What it does	Command
2. Data	Maps accent labels, filters and samples clips, writes the data card	<pre>python src/prepare_dataset.py --cv-dir <corpus>/en</pre>
3. Transcribe	Runs each model in Colab and saves transcripts as it goes	<pre>notebooks/03_transcribe.ip ynb</pre>
3. Score	Normalizes text and computes WER per clip	<pre>python src/score.py</pre>
4. Analyze	Bootstrap intervals, gap statistics, charts and worst-clips list	<pre>python src/analyze.py</pre>

The pipeline code was written with the help of an AI coding assistant (Claude). The research question, the decisions recorded in the data card, the Colab runs and the interpretation are part of this project's documented work.

Appendix: results for every model

Word error rate by accent group, every model (95% CIs; x-axes differ per panel)

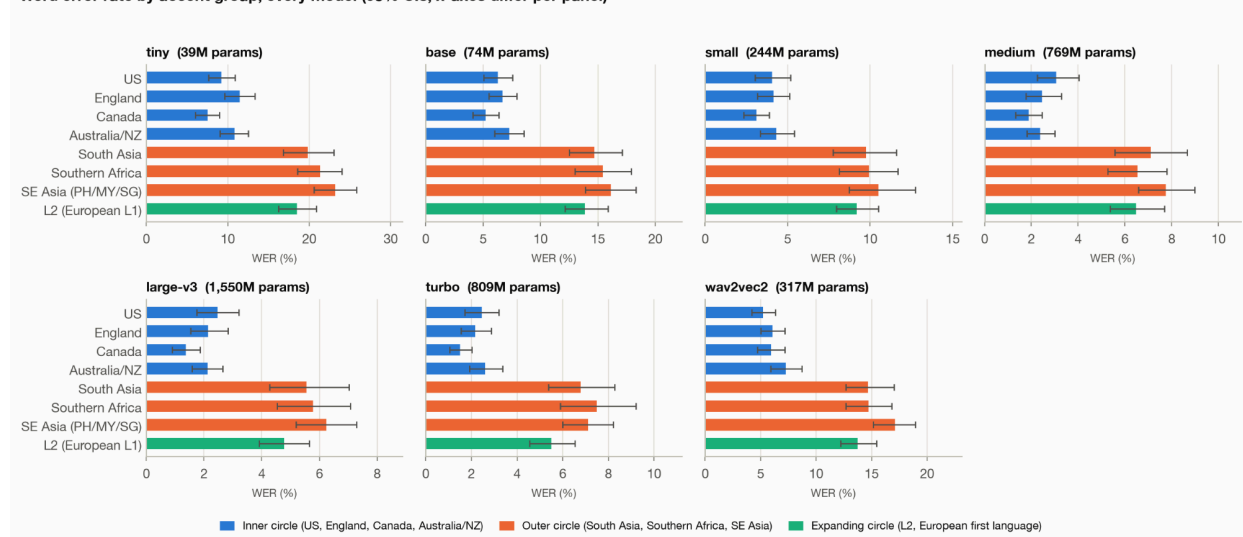


Figure A1. WER by accent group for all seven models, with 95% confidence intervals. Each panel has its own horizontal scale.

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